KNN and Bagging, Naive Bayes, Perceptron and Bayes' Rule

Machine Learning, Homework 3

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Notebook (`employee_attrition.ipynb`). The data are 902 employees from the IBM HR Analytics attrition set (35 columns). I explored income against years at the company and by department: R&D has the lowest average monthly income (6074) and Sales the highest (6817). I then label-encoded the categorical columns, dropped the three constant columns, made a 70/30 split and standardised the features. I wrote a KNN classifier from scratch and swept $K \in \{1, 3, \ldots, 15\}$. The best was $K = 9$ with test accuracy 0.830. Its recall on the minority class (left the company) is only 0.20, because 216 of the 271 test employees stayed. Bagging 9-NN models gave accuracy 0.827 with the same weakness (Figure 1).

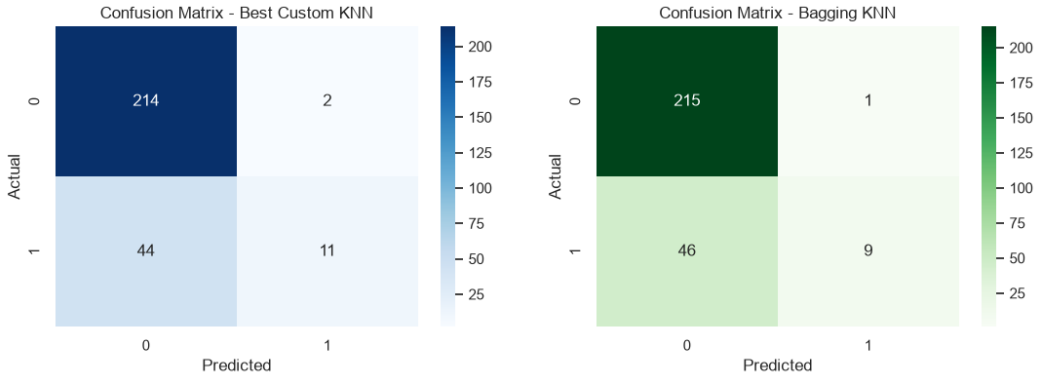


Figure 1: Confusion matrices of the from-scratch 9-NN (left) and bagged KNN (right).

Written part.

Naive Bayes with zero counts. I built four training examples where $P(A{=}1 \mid C{=}0) = 0$ and $P(B{=}1 \mid C{=}1) = 0$. A test case with $A = B = 1$ then has zero likelihood under both classes, so normalising the posterior divides by zero. That observation is inconsistent with the model; Laplace smoothing avoids it.

Perceptron by hand. From $w = (0, 0)$, $b = 0$, $\eta = 1$, only the first sample $(1, 2)$ triggers an update. This gives $w = (1, 2)$ and $b = 1$, after which all four points are classified correctly. The data are linearly separable (for example by $x_1 + x_2 = 0$), so the second epoch makes no update. I also explained why the perceptron never settles when the data are not linearly separable.

Rare-disease test. With prevalence 0.5%, sensitivity 90% and specificity 95%: $P(\text{disease} \mid +) = \frac{0.9 \cdot 0.005}{0.9 \cdot 0.005 + 0.05 \cdot 0.995} \approx 8.3\%$. With perfect sensitivity this only rises to $\approx 9.1\%$, because false positives from the large healthy population dominate. The test alone is not enough; symptoms, history and a confirmatory test are needed.